

WHITEPAPER

BLOCKCHAIN TECHNOLOGIES IN FINANCIAL MARKETS



 NOBLE

 OKCOIN

 quantifi

Contents

Executive Summary	3
Introducing Blockchain and Distributed Ledger	4
Distributed Ledger	4
Blockchain	5
Taxonomy of Distributed Ledger Platforms	7
Introducing Smart Contract	8
Ethereum	10
IBM Hyperledger	11
R3 Corda	12
Closing Thoughts	13
Appendix	14
Block Hashing Algorithm	14
Keys, Wallet and Transaction	15
References	17

About the Authors



Huang Zeng

Director, Client Services, EMEA/APAC, Quantifi

Huang is Director of Client Services, EMEA. He is responsible for coordinating system implementations, customisations, business consultancy as well as client and pre-sales support. Huang has extensive experience of enterprise solutions in the market and credit risk space, having worked for 6 years implementing and customising systems to meet evolving market and regulatory demands. Huang holds a B.S. in Natural Sciences from the University of Cambridge and a PhD in Physics from Imperial College London.



Che Guan

VP of Quantitative Strategies, Noble Markets

Che currently serves as a Vice President of Quantitative Strategies at Noble Capital International. Prior to that, he worked at JP Morgan and Dun & Bradstreet, responsible for data science solutions across retail and wholesale businesses as well as credit rating R&D. Che received a Ph.D. from University of Connecticut. His research interests include machine learning, big data analysis, cryptocurrency and blockchain technology with application to foreign exchange and commodity market, e.g. market microstructure research, asset allocation optimisation, etc.



Wilfred Daye

Head of Financial Markets, OKCoin

Wilfred is currently the Head of Financial Markets at OKCoin, a leading global blockchain technology company. Previously, he was the President and Senior Portfolio Manager for Noble Capital International. He was a structured credit trader and desk quant for UBS, Deutsche Bank, D.B. Zwirn and Barclays Capital. He began his career at Lehman Brothers and holds Series 7, 63, and 24 FINRA licenses.



Sachvir Cheema

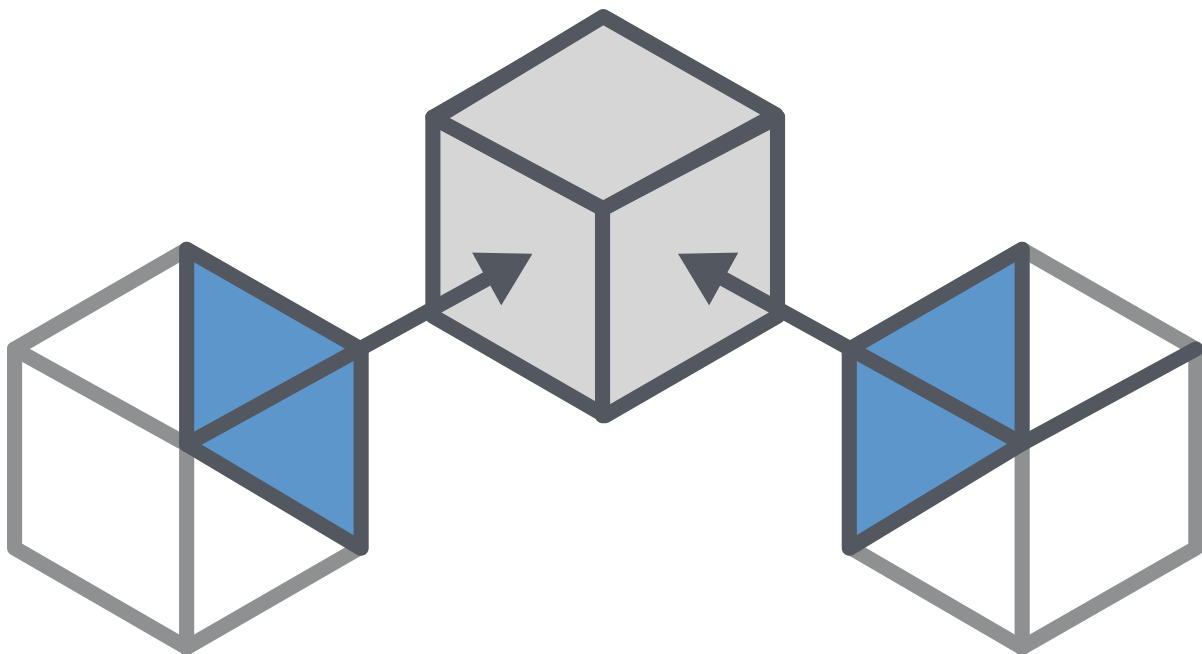
Director, Marketing & Business Development, Quantifi

Sachvir joined Quantifi in 2010. As Director of Marketing & Business Development, Sachvir is responsible for marketing communications, brand strategy development and implementation. Sachvir also manages all analyst relations and business development initiatives. Sachvir has over 15 years' experience in marketing and communications within financial services. Sachvir holds a BA (Hons) specialised in Marketing and Communications.

Executive Summary

Over the last 20 years, there have been significant technology advancements in the financial markets. Most recently, the interest in blockchain has been huge. The blockchain technology platform is a distributed ledger technology system that allows multiple parties to have simultaneous access to a constantly updated ledger that keeps a record of all transactions between participants. The block of transactions make up the chain, hence the name blockchain. This new technology has the potential to redefine operations and improve the efficiency, security and economics of the financial industry. It is time for financial institutions to change and benefit from the possibilities of distributed technologies.

Firms are investing millions in developing blockchain applications. Few now doubt that blockchain could soon revolutionise the global financial services industry.



Introducing Blockchain & Distributed Ledger

Distributed Ledger

In its simplest form, a distributed ledger is a digital record of transactions, shared instantaneously across a network of participants. The ledger may be used to register any transactions involving the exchange of something valuable, such as rights to payment or ownership of property. It is “distributed” because the transaction record is held by each of the users of the network and each user’s copy is updated with new information simultaneously.

The term “ledger” is a familiar concept in everyday banking. Bank statements set out the history of transactions on an account and summarise any outstanding amount owed by either the bank or the customer. In a distributed ledger, the accuracy of the database is confirmed by reconciling each individual version against all existing copies within a computer network. A “consensus” is reached to establish the true record, thereby avoiding the need for duplication as well as expensive and time-consuming reconciliation. This consensus process allows parties to immediately identify any instances of unauthorised tampering. Due to the distributed nature of the ledger, by construction, the distributed ledgers are not vulnerable to a single point of failure.

Blockchain is a distributed ledger that records digital transactions in a secure, transparent, immutable and auditable manner, without necessarily using a trusted third-party intermediary to perform these transactions. Here we illustrate a few selected characteristics:



Digital Distribution

A blockchain distributes data recording and transaction execution across the different computing nodes within a given network. There is no central authority, such as a central bank or clearinghouse, to act as a central data warehouse and intermediary.



Cryptographic Algorithm

Blockchain uses highly sophisticated and mature cryptography to ensure the reliability and security of data and transactions across the different computing nodes and information sharing amongst network participants.



Consensus

To validate information and accept new transactions, blockchain requires some or all of its nodes to reach a consensus. This consensus, which varies amongst different blockchain protocols, eliminates the need for a central authority to confirm and maintain a ledger of all transactions.



Immutability

Blockchain bundles transactions into “blocks”. Each block contains the previous block’s “hash” or digital signature, so that each block is linked to the one prior to it and, together, they interlock to form a chain. It is extremely difficult to change one block without changing the others that preceded it, making each block and the data it contains immutable. This immutability property helps to create the trust necessary for different parties to conduct business safely over the blockchain network.



Time-stamps

Blockchain time-stamps every new transaction or data entry in its block header, thereby allowing easy tracking and verification of information.



Resilience

A distributed database is, by its nature, more resistant to accidental failures or malicious attacks than a centralised system. Specifically, if one or more nodes on a blockchain fail or are hacked, the rest of the system can still function reliably.



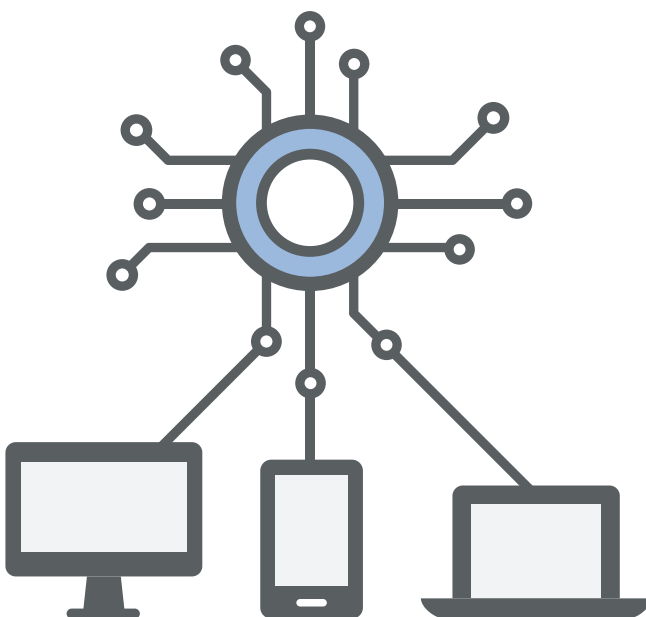
Security

Under the distributed nature of blockchain, a hacker can only corrupt a blockchain by initiating a 51% attack on the network. Encryption and multi-step verification procedures add protection, as do time-stamped transactions, data that cannot be altered without trace and the need for consensus to accept transactions.

Blockchain

A blockchain is a technical component of a distributed ledger and refers to the chain of transactions that reside within the ledger. Transactions are grouped into “blocks” and as they are verified, a new “block” is added to the chain of previous transactions. The ledger is updated instantaneously, permanently and irrevocably for all users to reflect the new status of the ledger with the additional new block. Therefore, the blockchain is an accurate record of the history of the entire ledger. Not all distributed ledgers use blockchain technology, though the terms are often used interchangeably.

Bitcoin, invented by Satoshi Nakamoto [1], is a digital asset and a peer-to-peer decentralised payment system. It was created as a reward for payment processing work in which participants can offer their computing power to verify and record payments into a public ledger. Such activity was rewarded with transaction fees and the creation of Bitcoins [2]. Bitcoin is the first generation of blockchain, a public ledger consisting of blocks that holds time-stamped batches of recent, valid transactions. In a nutshell, Bitcoin has two main components: miners that solve computational problems using special-purpose chips through proof-of-work and full nodes that store the blockchain. These nodes relay and validate transactions and blocks that miners build. Figure 1 gives a simple example. For details, refer to the Appendix.



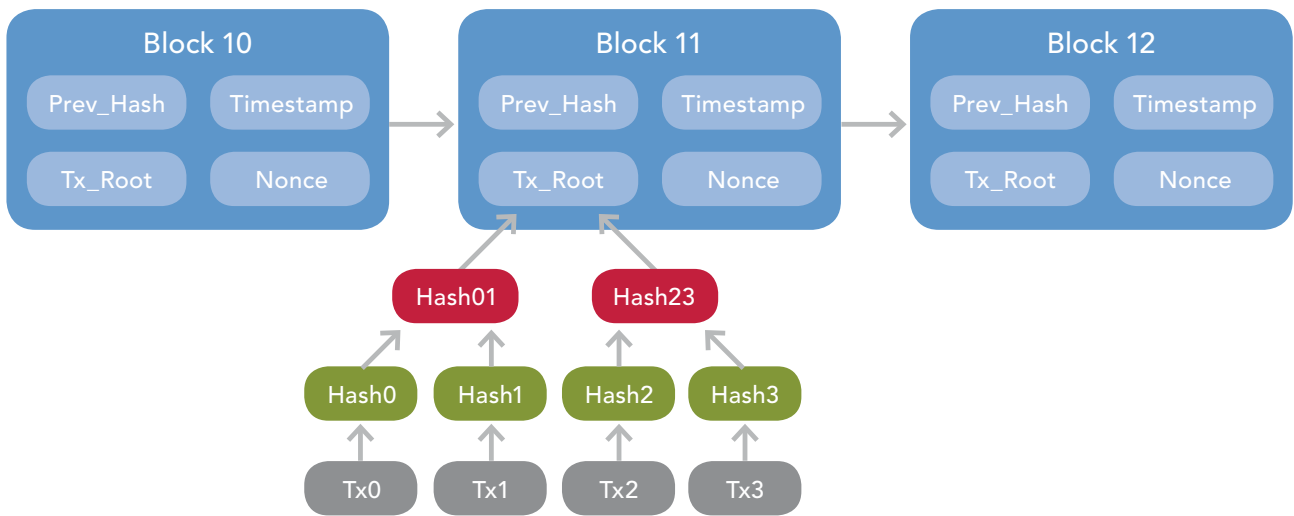


Figure 1. An Example of Blockchain [3]

Blockchain is generated through mining, which is a record-keeping service. To be accepted by the rest of the network, a new block must contain a “Proof-of-Work” (PoW). The PoW requires miners to find a number referred to as a nonce, such that when the block content is hashed along with the nonce, the result is numerically smaller than the network’s difficulty target (difficulty is a measure of how difficult it is to find a hash below a given target). This proof is easy for any node in the network to verify, but extremely time-consuming to generate (i.e. for a secure cryptographic hash, miners must try many different nonce values before meeting the difficulty target) [3]. Furthermore, the PoW system, alongside the chaining of blocks, makes modifications of the blockchain extremely hard, as an attacker must modify all subsequent blocks in order for the modifications of one block to be accepted. As new blocks are mined, the difficulty of modifying a block increases as time passes, and the number of subsequent blocks increases. In addition, an attacker can only try to change one of the attacker’s own transactions (relate to attacker’s own private key) to take back money this attacker recently spent [1], [3], [4].

A successful miner finding a new block is rewarded with newly created Bitcoins and transaction fees. Paying a transaction fee is optional, but may speed up confirmation of the transaction. To store the digital credentials for Bitcoin holdings and allow the owner to access and spend them, a wallet is needed. Bitcoin uses public-key cryptography, in which two cryptographic keys, one private and one public, are generated. Therefore, a Bitcoin digital wallet is a collection of these keys. Bitcoins are not tied to real-world entities, but rather Bitcoin addresses; owners of Bitcoin addresses are not explicitly identified, but all transactions on the blockchain are public [2]. The Appendix provides additional details of the technology.

Taxonomy of Distributed Ledger Platforms

In Figure 2, we categorise distributed ledger platforms into two types: “permissionless” and “permissioned.”

Permissionless blockchains are public and open for anyone to read, write transactions to and participate in the consensus process. Key benefits of permissionless blockchains include: complete disintermediation, as entry costs for new participants are minimal, and that no participant has exclusive control over the blockchain network. However, there are limits to the potential use of permissionless blockchains within financial services. The relatively slow transaction speed is one such issue, caused by the fact that these blockchains continue to grow because all transactions are included on the chain. Notably, Bitcoin can only execute a theoretical maximum of seven transactions per second. Also, permissionless blockchains provide limited data privacy. These limitations are serious problems for most financial services applications (i.e. instances

where participants in a transaction do not want information to be visible to others). For example, the permissionless, public shared ledgers are protocols associated with Bitcoin, Litecoin and Ethereum.

In general, the financial services industry likes the ability to keep client and transaction information private, to grant different access levels to different parties, to scale up transactions and databases and to modify the consensus structure if members agree. Permissioned blockchains only permit consortium members access to information and transaction history. To verify data and transactions, a subset of the nodes, as determined by the consortium, can reach a consensus based on pre-defined criteria. Further, participants in the consortium can agree to change the criteria for verifying and recording transactions. For example, the permissioned, private shared ledgers are Corda and Hyperledger. The permissioned ledgers can also be shared with everyone for read access, such as Ripple and Everledger.

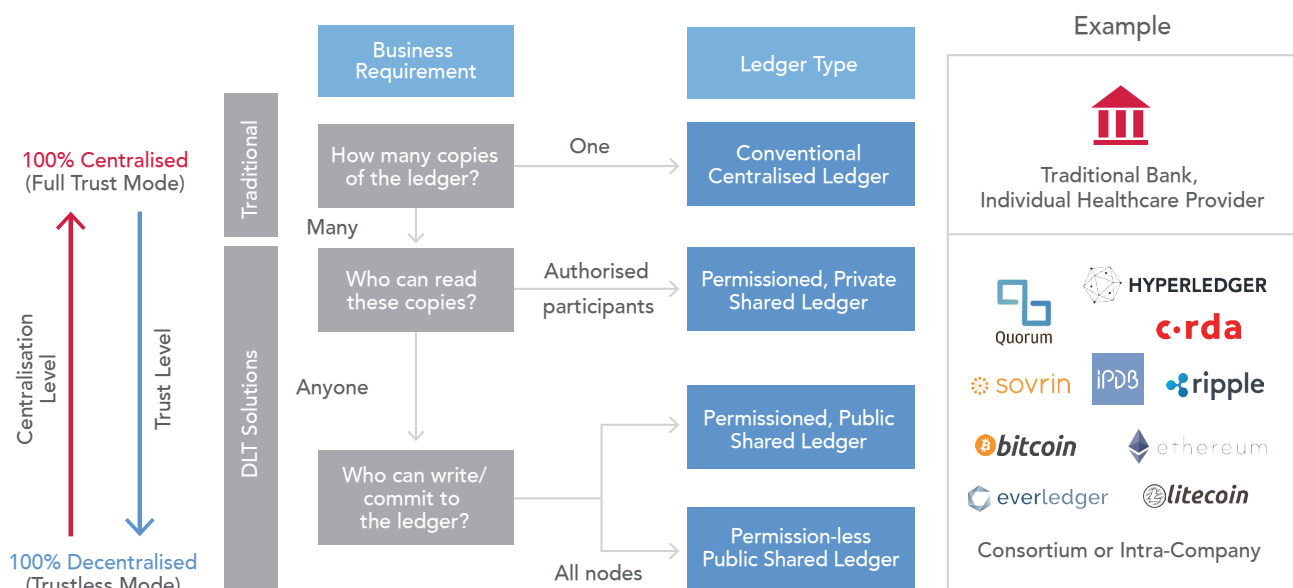


Figure 2. Permissioned vs Permissionless Distributed Ledger Technology

Introducing Smart Contract

Many blockchains are taking advantage of “smart” contracts. In 1996, the computer scientist and legal scholar Nick Szabo described a smart contract as “a set of promises, specified in digital form, including protocols within which the parties perform on these promises” [5]. A simpler way of looking at a smart contract is as a logical if-then statement, i.e. if a condition is met, then a result is executed.

A traditional contract is a promise between two or more parties that are legally binding. For instance, Alice promises to pay Bob money in return for use of Bob’s house. In the tenet of distributed ledger technologies, a “smart” contract is one where conditions are both evaluated and executed by computer codes making the transaction trustless. The “smart” rental contract between Alice and Bob would lock Alice out of her apartment if she fails to pay.

“A smart contract is a computerised transaction protocol that executes the terms of a contract. The general objectives are to satisfy common contractual conditions (such as payment terms, liens, confidentiality and even enforcement), minimise exceptions, both malicious and accidental, and minimise the need for trusted intermediaries. Related economic goals include: lowering fraud loss, arbitrations and enforcement costs and other transaction costs” [5].

Once participants agree to terms, conditions, and outcomes, the smart contract is coded and recorded on a distributed ledger. The code typically contains references to external data sources that the smart contract needs to work. Once a smart contract is recorded, it cannot be modified without the participating parties’ permission, therefore, ensuring the fidelity of contractual terms. Current smart contract development efforts are mostly focused on three areas: security, accessibility and legal certainty.

Smart contracts are digital contracts allowing terms contingent on decentralised consensus that are self-enforcing and tamper-proof through automated execution. Contingencies, such as allocation of property and control rights in a smart contract, should be codified to automate execution and reduce cost. By enabling the use of smart contracts, blockchain technology can increase contractibility and enforceability in contingent contracts that facilitate exchanging money, property, shares, service or anything of value in an algorithmically automated and conflict-free way.

There are three popular blockchains using smart contracts: Ethereum, IBM Hyperledger Fabric and R3 Corda. These are good examples of distributed ledger platforms as described in the previous section. Table 1 gives a comparison amongst the three blockchains. At a high level, Ethereum is a permissionless public blockchain as the participation is permissionless. On the other hand, Hyperledger Fabric and R3 Corda are permissioned as participants are selected in advance and access to the network is restricted. Details about Ethereum, Hyperledger and Corda as well as their comparisons will be explained in the following sections.

	Ethereum	Hyperledger Fabric	R3 Corda
Description of platform	- Generic blockchain platform	- Modular blockchain platform	- Specialised distributed ledger platform for the financial industry
Governance	- Ethereum developers	- Linux Foundation	- R3
Mode of operation	- Permissionless, public or private	- Permissioned private	- Permissioned, private
Consensus	- Mining based on proof-of-work (PoW) - Ledger level	- Broad understanding of consensus that allows multiple approaches - Transaction level	- Specific understanding of consensus (e.g. notary notes) - Transaction level
Smart contracts	- Smart contract code (e.g. Solidity)	- Smart contract code (e.g. Go, Java)	- Smart contract code (e.g. Kotlin, Java) - Smart legal contract (legal prose)
Currency	- Ether - Tokens via smart contract	- None - Currency and tokens via chaincode	- None

Table 1. Comparison of Ethereum (Permissionless), Hyperledger Fabric and Corda (Permissioned) [6]

Ethereum

Ethereum is a decentralised platform that runs smart contracts. It uses a blockchain to synchronise and store the system state along with a cryptocurrency called 'ether' to meter and constrain execution resource cost [7-9]. From a computer science perspective, Ethereum is a deterministic, but practically unbounded state machine with two basic functions: a globally accessible singleton state and a virtual machine that applies changes to that state [9]. It quickly becomes a platform for programming Decentralised Applications (DApps) - a web application built on top of open, decentralised, peer-to-peer infrastructure services. DApps enable developers to build powerful applications as well as reduce or eliminate censorship, third-party interface and counterparty risk [7]. The Ethereum project was developed by the Ethereum Foundation via an ether pre-sale in August 2014.

Ethereum shares many common elements with other open blockchains (e.g. bitcoin blockchain): a peer-to-peer network connecting participants, a consensus algorithm for synchronisation of state (PoW), a digital currency (Ether) and a global ledger (the Blockchain) [9]. From a practical perspective, the bitcoin blockchain is a distributed consensus state machine, which tracks the state of units of bitcoin and their ownership. Ethereum is also a distributed state machine. But instead of tracking only the state of currency ownership, it tracks the state transitions of a general-purpose code and data store. For example, it can load code into its state machine and run that code, storing the resulting state changes in the blockchain. Ethereum state changes are governed by the rules of consensus, and the state is distributed globally on a shared ledger. Furthermore, unlike bitcoin blockchain, which is a limited scripting language, Ethereum is designed to be a general purpose, programmable blockchain that runs a virtual machine capable of executing code of arbitrary and unbounded complexity [9].



IBM Hyperledger

Hyperledger is an umbrella project of open source blockchains [10]. As depicted in Figure 3, there are five frameworks beneath the umbrella [11], including:

To facilitate these frameworks, there are five major modules to facilitate development [11], including:

- | | |
|---|--|
| 1 Hyperledger Fabric: designed as a foundation for developing applications or solutions with a modular architecture; | 1 Hyperledger Caliper: a blockchain benchmark tool for measuring the performance of a specific blockchain implementation; |
| 2 Hyperledger Sawtooth: a modular platform for building, deploying and running distributed ledgers; | 2 Hyperledger Cello: brings the on-demand "as-a-service" deployment model to the blockchain ecosystem; |
| 3 Hyperledger Iroha: designed to incorporate into infrastructural projects requiring distributed ledger technology; | 3 Hyperledger Composer: collaboration tool for building blockchain business networks. Accelerates development and deployment of smart contracts; |
| 4 Hyperledger Indy: purpose-built for decentralised identity; | 4 Hyperledger Explorer: can view, invoke, deploy or query blockchain information; |
| 5 Hyperledger Burrow: a permissionable smart contract node that handles transactions and executes smart contract code | 5 Hyperledger Quilt: offers interoperability between ledger systems. |

Hyperledger Fabric is the first business blockchain framework contributed by IBM and Digital Asset. It is often used to compare to other blockchains. Similar to Ethereum, Hyperledger Fabric is also highly flexible. Ethereum's smart contract engine makes it a popular platform for many applications. However, its permissionless mode and total transparency comes at the cost of performance scalability and privacy. Fabric solves performance scalability and privacy issues by permissioned mode and, specifically, by using a Byzantine fault tolerance algorithm and fine-grained access control. Furthermore, the modular architecture allows Fabric to be customised to a multitude of applications [6].

Infrastructure

Technical, Legal,
Marketing, Organisational

Ecosystems that accelerate open
development and commercial adoptions



CloudFoundry

Node.js

Hyperledger

Open Container
Initiative

Frameworks

Meaningfully differentiated approaches
to business blockchain frameworks
developed by a growing community
of communities from the entire industry

Hyperledger
Sawtooth

Hyperledger
Iroha

Hyperledger
Fabric

Hyperledger
Indy

Hyperledger
Burrow

Modules

Typically built for one framework and,
through common license and the
community of communities approach,
ported to other frameworks

Hyperledger
Cello

Hyperledger
Chaintool

Hyperledger
Explorer

Hyperledger
Composer

Figure 3: Hyperledger Modular Umbrella Approach [12]

R3 Corda

Corda is an enterprise-grade blockchain platform developed by a distributed database technology company called R3 [13]. It was originally built for recording and processing financial agreements. Corda enables institutions to transact directly using smart contracts, whilst ensuring high levels of privacy and security [14]. Similar to Ethereum, Corda runs inside a relatively powerful virtual machine and can contain complex logic. However, the term “contract” in Ethereum refers to an instantiation (like an object in an Object-Oriented program) of a program that is replicated and maintained by every participating node. In contrast, implementation of the smart contract in Corda refers to a set of functions, only one of which is a part of keeping the system synchronised (the verify function which is pure and stateless) [15].

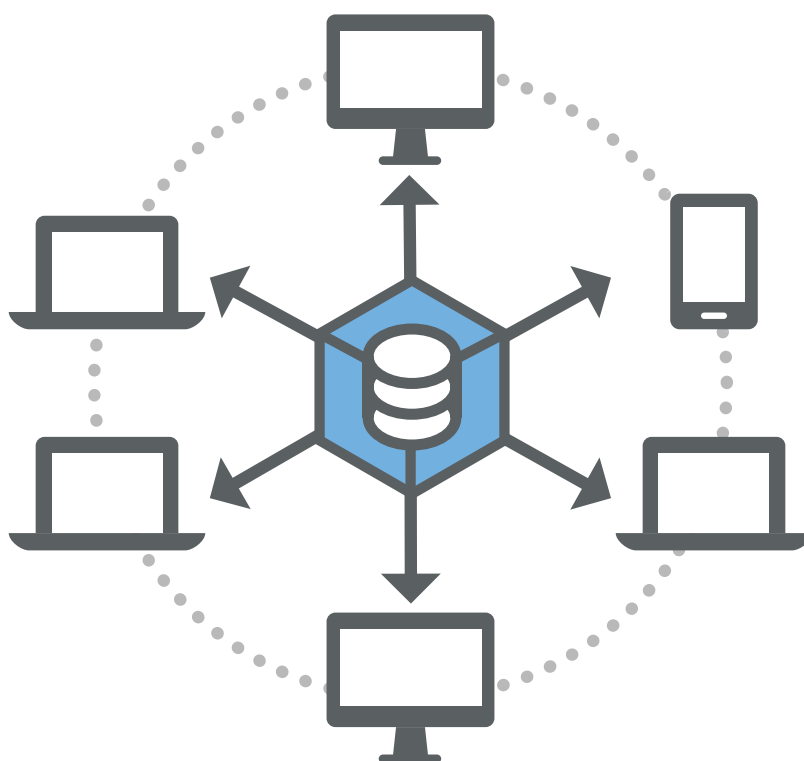
Similar to Hyperledger Fabric, consensus in Corda is reached at transaction level by involving parties only. However, Corda has been consciously designed for the financial services industry. Most notably, it takes the highly regulated environment into account by augmenting smart contracts with legal prose. Its sole focus on financial services transactions simplified its architectural design and can be seen as a complement to Fabric [6].

Closing Thoughts

Distributed ledger technology (DLT) has gained a lot of attention lately. It has caught the attention of many firms, including those active in payment, clearing and settlement, for its potential to disrupt financial services by making them more efficient, secure, transparent and reliable. DLT could bring a number of benefits to the market, notably enhanced reporting and data management capabilities, more efficient and streamlined post-trade processes and reduced costs.

Blockchain is the most recognised form of distributed ledger. Blockchain based DLT, which was first applied as the underlying technology for virtual currencies such as Bitcoin, has a number of possible applications within finance. For example, DLT could be applied to digital record keeping, smart contracts, digital currencies and digital assets.

DLT represents a promising source of future innovation in financial markets as it is starting to prove itself beyond just cryptocurrency. The opportunities arising from blockchain are vast, there are however many challenges to contend with before the full potential of the technology can be realised. It will take few years for DLT to make its way into the mainstream, however, there is no doubt that it will have a significant impact on the global financial markets.



Appendix

Block Hashing Algorithm

As a data container, a block consists of a header (containing metadata) with 80 bytes in average, and transactions with at least 250 bytes in average. Block size and transaction counter are also in a block. The average block contains more than 500 transactions. As described in Table 2, a block header consists of three sets of block metadata: a reference to a previous block hash, a set of mining competition strings, which includes the difficulty, timestamp, and nonce and a Merkle tree root summarising all the transactions in the block [16]. To mine bitcoins, the Proof of Work hashcash function is used. The PoW algorithm repeatedly hashes the block header whilst incrementing the counter and extraNonce fields. A solo miner increments nonce until it overflows, which then increments extraNonce and resets nonce. Incrementing the extraNonce field entails recomputing the Merkle tree, as explained in the next paragraph.

Field	Purpose	Updated When...	Size (Bytes)
Version	Block version number	You upgrade the software and it specifies a new version	4
Governance	256-bit hash of the previous block header	A new block comes in	32
Mode of operation	256-bit hash based on all the transactions in the block	A transaction is accepted	32
Consensus	Current timestamp as seconds since 1970-01-01T00:00 UTC	Every few seconds	4
Smart contracts	Current target in compact format	The difficulty is adjusted	4
Currency	32-bit number (starts at 0)	A hash is tried (increments)	4

Table 2. Contents in a Block Header [17]

To further understand the hash of all transactions in the block, it is necessary to understand the mechanism of a Merkle tree. A Merkle tree is constructed by recursively hashing pairs of nodes until there is only one hash, known as the root or Merkle root. Rather than directly storing transactions in the Merkle tree, transactions are hashed, and the resulting hash is stored in each leaf node. For example, and as depicted in Figure 4, Hash 0 is the result of hashing the result of concatenating Hash 0-0 and Hash 0-1. That is, $\text{Hash } 0 = \text{Hash}(\text{Hash } 0-0 + \text{Hash } 0-1)$, where the symbol “+” denotes concatenation and the hash algorithm used is SHA256, which is applied twice (double-SHA256). Finally, a top hash is produced in the top of a hash tree. Before downloading a file on a peer-2-peer network, in most cases, the top hash is acquired from a trusted source. When the top hash is available, the hash tree can be received from any non-trusted source, like any peer in the peer-2-peer network. Then, the received hash tree is checked against the trusted top hash and, if the hash tree is damaged or fake, another hash tree from another source will be tried until the program finds one that matches the top hash [18].

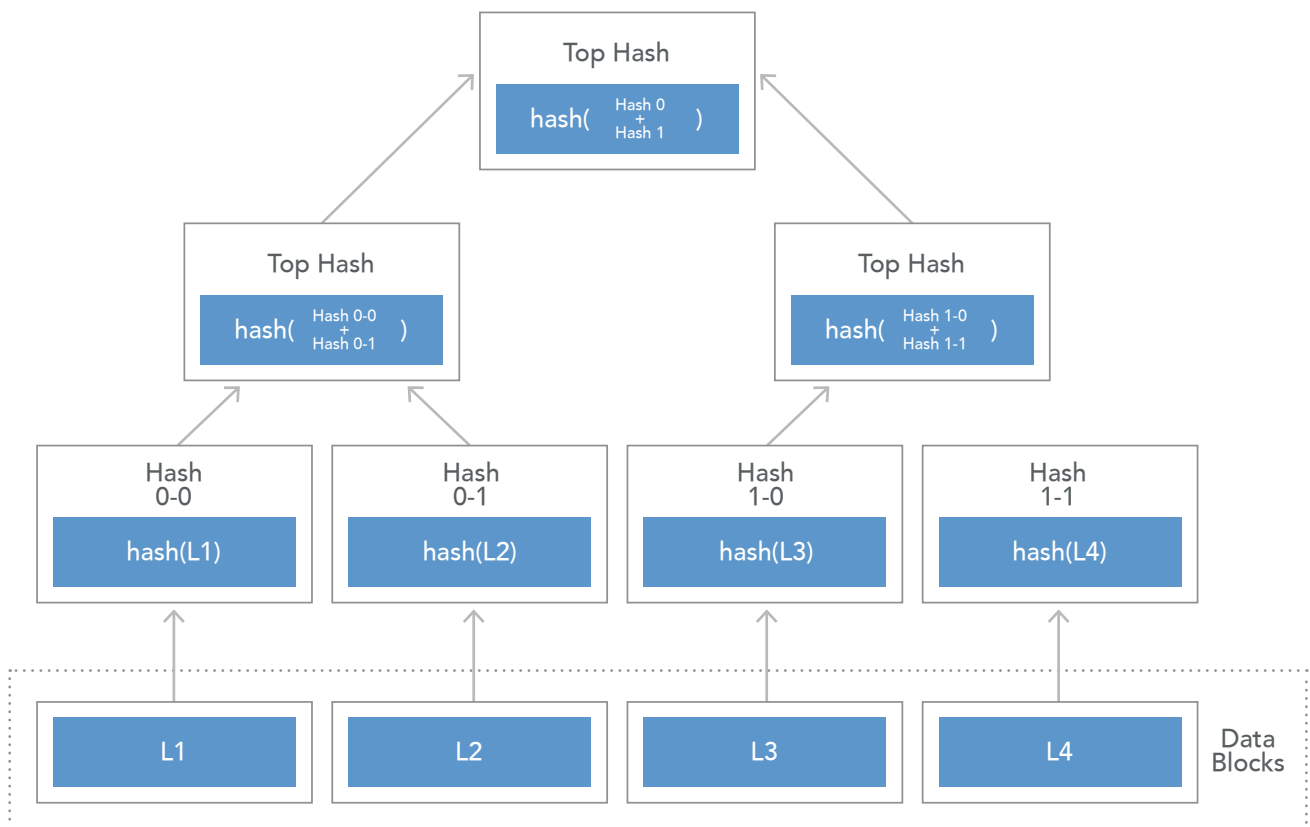
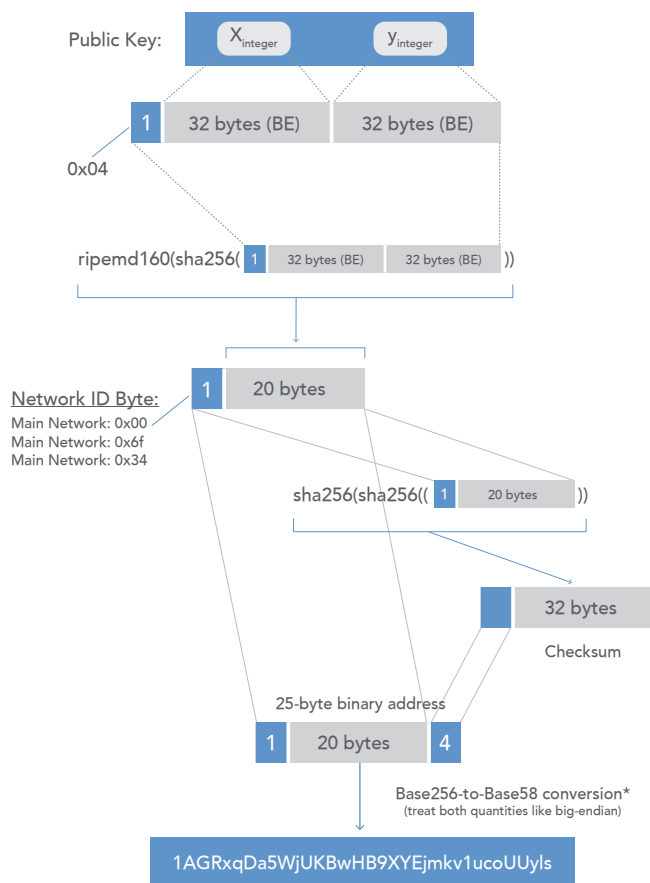


Figure 4. An Example of a Binary Hash Tree [18]

Keys, Wallet and Transaction

Ownership of Bitcoin is established through private and public digital keys as well as Bitcoin addresses stored in the wallet. The private key is like the secret account PIN, the public key is like a bank account number and the bitcoin address is like the beneficiary name on a check. A wallet is a container for keys, not coins (coins are stored on the blockchain). To transfer values between wallets in the bitcoin system, transactions serve as the most important data structure [20].

Keys and addresses are sequentially generated. The private key is a number, which is like picking a number between 1 and 2256. From the private key, the elliptic curve multiplication, a one-way cryptographic function is used to generate a public key. Next, to generate a Bitcoin address, as shown in Figure 5, the generated public key is put through several SHA-256 and RIPEMD-160 operations, until eventually being converted into a format called Base-58 (it is an encoding that removes the possibility of similar looking characters). Finally, an identifying number is added to the beginning of the address, indicating it is a public Bitcoin network address [19].



1. Having a private ECDSA (Elliptic Curve Digital Signature Algorithm) key
2. Take the corresponding public key generated with it (33 bytes, 1 byte 0x03 and 32 bytes corresponding to X coordinate).
3. Perform SHA-256 hashing on the public key.
4. Perform RIPEMD-160 hashing on the result of SHA-256.
5. Add version byte in front of RIPEMD-160 hash (0x00 for Main Network).
6. Perform SHA-256 hash on the extended RIPEMD-160 result.
7. Perform SHA-256 hash on the previous SHA-256 hash.
8. Take the first 4 bytes of the second SHA-256 hash. This is the address checksum.
9. Add the 4 checksum bytes from stage 7 at the end of the extended RIPEMD-160 hash from stage 4. This is the 25-byte binary Bitcoin Address.
10. Convert the result from a byte string into a base58 string using Base58Check encoding. This is the most commonly used Bitcoin Address format.

Figure 5. Elliptic Curve Public Key to BTC Address Conversion [19]

To manage keys and addresses, the 'wallet', a data structure, is developed. There are two primary types of wallets: nondeterministic and deterministic wallets. The former is a collection of randomly generated keys, whereas the latter contains a deterministic sequence of keys derived from a seed (a single master key). The most advanced form of deterministic wallets is a hierarchical deterministic (HD) wallet, which contain keys derived in a tree structure. It offers two major advantages over the nondeterministic: it expresses organisational meaning, and users can create a sequence of public keys without gaining access to the corresponding private keys [16].

A wallet application calculates the bitcoin "balance" through taking the sum of all spendable unspent transaction outputs (UTXOs). To understand value transferring and validation process of a transaction, the important step is to understand transaction outputs, inputs and their relationship. Transaction

outputs consist of a bitcoin amount and a locking script which is a spending condition placed on an output. Many transactions spend outputs locked with a Pay-to-Public-Key-Hash (P2PKH) script, and the outputs can be unlocked by using a public key and a digital signature (produced by the Elliptic Curve Digital Signature algorithm) created by the corresponding private key. On the other side, transaction inputs consist of a transaction ID, an output index (identifying UTXO referenced in the transaction) and an unlocking script which satisfies the conditions placed on an output by the locking script. A validation software validates transactions by executing the locking and unlocking scripts together. To offer advanced functions for corporate needs, scripts like Multisignature (N public keys are recorded in the script and at least M of those must provide signatures to unlock the funds) and Pay-to-Script-Hash (P2SH) script with timelocks create transactions with complex conditions (e.g. corporate governance control, theft or loss protection, etc).

References

- [1] Satoshi Nakamoto, "Bitcoin a peer-to-peer electronic cash system," 2009
- [2] Bitcoin, Wikipedia, <https://en.wikipedia.org/wiki/Bitcoin>
- [3] Matthaus Wander, "How Bitcoin Works," https://www.vs.uni-due.de/wander/20110629_Bitcoin_Wander.pdf
- [4] Bitcoin, <https://www.bitcoin.com>
- [5] Nick Szabo, "The Idea of Smart Contracts," 1997
- [6] Martin Valenta, Philipp Sandner, "Comparison of Ethereum, Hyperledger Fabric and Corda," FSBC Working Paper, June 2017
- [7] Ethereum Project, <https://ethereum.org/>
- [8] Ethereum, Wikipedia, <https://en.wikipedia.org/wiki/Ethereum>
- [9] Andreas M. Antonopoulos and Gavin Wood "Mastering Ethereum," O'Reilly, December 2018
- [10] Hyperledger, Wikipedia, <https://en.wikipedia.org/wiki/Hyperledger>
- [11] Hyperledger Official Website, <https://www.hyperledger.org/projects>
- [12] Arnaud Le Hors, "Hyperledger Community Update," February 2018
- [13] R3, Wikipedia, [https://en.wikipedia.org/wiki/R3_\(company\)](https://en.wikipedia.org/wiki/R3_(company))
- [14] R3, Official Website, <https://www.r3.com/technology/>
- [15] Richard G. Brown, James Carlyle, Ian Grigg, Mike Hearn, "Corda: An Introduction," corda.net, 2016
- [16] Andreas M. Antonopoulos, "Mastering bitcoin – Programming the Open Blockchain, 2nd Edition," O'Reilly Media, June 2017
- [17] Blockchain Hashing Algorithm, Wikipedia, https://en.bitcoin.it/wiki/Block_hashing_algorithm
- [18] Merkle Tree, Wikipedia, https://en.wikipedia.org/wiki/Merkle_tree
- [19] Technical background of version 1 Bitcoin Address, Wikipedia, https://en.bitcoin.it/wiki/Technical_background_of_version_1_Bitcoin_addresses
- [20] BA.et Bitcoin, "Bitcoin Serverless Wallet and Vault," pp 182-184



This new technology has the potential to redefine operations and improve the efficiency, security and economics of the financial industry. It is time to for financial institutions to change and benefit from the possibilities of distributed technologies.



Quantifi is a provider of risk, analytics and trading solutions. Our award-winning suite of integrated pre and post-trade solutions allows market participants to better value, trade and risk manage their exposures and responds more effectively to changing market conditions. Quantifi is trusted by the world's most sophisticated financial institutions, including five of the six largest global banks, two of the three largest asset managers, leading hedge funds, insurance companies, pension funds, and other financial institutions across 40 countries. Renowned for our client focus, depth of experience, and commitment to innovation, Quantifi is consistently first-to-market with intuitive, award-winning solutions.

enquire@quantifisolutions.com | www.quantifisolutions.com

EMEA +44 (0) 20 7248 3593 NA +1 (212) 784 6815 APAC +61 (02) 9221 0133

Follow us on   



The Noble group of companies ("Noble") was founded upon the idea that the barrier to innovation within financial services was simply too high. Noble's mission was to provide solutions that solve the problems around adopting innovative ideas for financial services. Our platform provides services that are cost-effective and is built with the digital age in mind. Our platform allows financial institutions to service its clients, allowing them to focus on their core competencies and create new product ideas for their clients without having to sacrifice the outcome due to legacy limitations that exist today. In 2015, Noble took steps to extend the reach of its platform and further its vision of providing innovative financial services to its clients by acquiring Noble Bank International.

www.noblebankint.com



Founded in 2013, OKCoin is a world leading digital asset trading platform. OKCoin currently provides fiat trading with major digital assets, including Bitcoin, Bitcoin Cash, Ethereum, Ethereum Classic, and Litecoin. OKCoin has recently expanded to the United States and is looking to serve more users around the world in the future.

www.okcoin.com